CURRENT NEWS AND NOTES.

THE SOCIETY OF MODEL ENGINEERS will hold a meeting and an exhibit of models at 1200 Madison Avenue on the evening of Feb. 2. The movement is headed by Mr. C. Kehrer at that address.

MR. B. A. BEHREND, formerly assistant chief electrician of the Oerlikon Engineering Works, Switzerland, has taken charge of the alternating-current department of the Bullock Electric Manufacturing Company.

TRAIN LIGHTING.—Consul-General F. H. Mason, at Berlin, reports to the State Department quite extensively on the recent railway accident and fire at Offenbach-on-Main, and the resultant agitation for a general adoption in Germany of electric lighting for trains.

TO RUN EVERYTHING.—The Canadian Independence League of Montreal, declaring for a Republic of Canada, demands that telegraphs, canals, railroads, etc., shall be owned by the State, and that municipalities shall own and operate the electrical plants, street railways, waterworks, gas works, telephones, and any other old thing. "A large order."

MR. STEINMETZ'S LABORATORY DESTROYED BY FIRE.—Charles Proteus Steinmetz met with a severe loss last week by the destruction of his private electro-chemical laboratory by fire, which probably had its origin in a coal stove. A considerable part of the apparatus was saved, but a number of interesting investigations being carried out in the laboratory are indefinitely delayed.

ELECTRICITY IN BRAZIL.—An excellent electrical review is now being published at Rio de Janeiro by the electrical and mechanical importing house of James Mitchell & Co., who, by the way, now employ 100 people. It is the first journal of the kind in Brazil, and doubtless in all South America, and is a laudable piece of technical enterprise. A member of the firm is Mr. Adolph Aschoff, who visited this country at the time of the Chicago Exposition, and was attached to the Brazilian Commission. His work and energy had much to do with the introduction of the electric railway into Brazil soon after.

WIRELESS IN 1869.—A wireless telegraph system, devised by Dr. Mahlon Loomis was, it appears, before the United States Senate in 1869, with the request for an appropriation of $50,000. The doctor lived in Washington and did not get the money. His petition said: "The nature of the discovery or invention, in general terms, consists in establishing an electrical current or circuit for telegraphing without the aid of wires or cables to form such electrical currents or circuits. As in dispensing with the double wire (which was at first used), and using but one wire, allowing and relying upon the earth to form one-half of the circuit, so now I propose to dispense with both wires and with all artificial batteries, and using the earth, as now, to form one-half of the circuit, and the continuous electrical element far above the surface of the earth for the other circuit."

TAXATION IN WISCONSIN.—The first report of the Wisconsin Tax Commission, created two years ago, recommends a continuance of the license fee system of collecting taxes from corporations, but would increase the rate to make the returns more nearly equal to what would be collected from the same properties on the ad valorem basis. For street railroads, telephone companies, and electric-power companies, the license fee recommended is a minimum of 3 per cent of the annual gross receipts, where these do not exceed $25,000, rising by one-tenth of 1 per cent for each additional $7 500 of receipts until $100,000 is reached, when the rate will be 4 per cent, and rising thence at the same rate for each additional $10,000 of receipts up to 5 per cent for receipts of $200,000 or more. For telegraph companies the ad valorem plan is advised.

VACUUM TUBE LIGHTING.—According to newspaper reports Mr. Tesla is again working upon vacuum tubes. In an authoritative statement in the New York *Sun* he says: "The lamps are glass tubes which may be bent in any ornamental way. I most generally use a rectangular spiral, containing about 20 ft. to 25 ft. of tubing, making some twelve to fourteen convolutions. The total illuminating surface of a lamp is from 300 to 400 square inches. The ends of the spiral tube are covered with a metallic coating and provided with hooks for hanging the lamp on the terminals of the source of oscillations. The tube contains gases rarefied to a certain degree, determined in the course of long experimentation as being conducive to the best results." After discussing the subject in general terms, he adds: "I have no difficulty now to illuminate the room with such wireless lamps, but a number of improvements must be made yet before it can be generally introduced." Wherein this system differs from that of Geissler, Moore and the several earlier ones of Tesla, is not stated.

ELECTRIC REDUCTION OF ORES.—A special dispatch from Kansas City of Jan. 29 says: Charles E. Finney, manager of M. Guggenheim's Sons of New York, the smelters; Charles S. Gleed of Topeka, a director of the Santa Fé Railroad, and Prof. Lucien Blake, head of the electrical department of the University of Kansas, met to-day to discuss a proposition that may produce most important results to smelters and owners of mines of precious metals. Prof. Blake has been experimenting for some time in his laboratory on an electrical process by which metal may be obtained from ore cheaper than by any process now in use. The purpose of the method is to increase the value of low-grade ore. If the proposition is carried out the test will be made in a Denver mill to be furnished by Mr. Gleed, while Mr. Finney supplies the ore. The laboratory tests already made have, it is claimed, gone far enough to show the possibility, if not the probability, of the use of electricity in the separation of metals from their ores. This method, if it proves successful, will be much cheaper than the methods now in use in smelters.

ELECTRIC POWER IN AFRICA.—At a meeting of Rezende, Limited, a Rhodesian company, recently held in London, the chairman said the Umtali Falls were near their mine, and from the outset it had been the directors' intention to use the power there available, but owing to conflicting claims they did not feel it safe to order the necessary machinery, but on the day that matters were put on a proper business footing with the chartered company they immediately proceeded to order the water power and electrical plant the specification and tenders for which had long before been obtained in anticipation. They had the right to half the power of the falls, and the plant, which was now in course of shipment, was calculated to deliver about 150 hp effective at the mine. Until the plant was erected and started they could not bring their additional 10 head of stamps into use. Two Pelton water wheels would drive a two-phase generator, which would produce current at 2500 volts. This current would be conveyed about 1½ miles to the mine, where it would be utilized in driving the motors, working the mill, rock breakers, cyanide works pump, mine pump, hoisting gear, rock drills, and other machinery, besides supplying electric light. Until this installation was complete they were dependent on wood and oil fuel. The plant ordered had now been practically all shipped, and some of the material was actually in Umtali.

ANOTHER TESLA DISCOVERY.—Through the daily newspapers of Wednesday of this week Tesla announced another new discovery in electricity, which he says will require a large part of technical literature to be rewritten. The discovery is that capacity is not fixed and unalterable. In the announcement Tesla says he has found that capacity increases as the conducting surface is elevated in open space, from one-half to three-fourths of one per cent per each foot of elevation, though in buildings or near large structures this increase often amounted to 50 per cent per foot of elevation. Moreover, he says he has observed that there is a diurnal variation of the capacity, with a maximum during the day and a minimum during the night; and the astonishing statement is made that while both the sun and moon cause a slight increase in capacity, in the former case it is due to sunlight, but in the latter case not due to moonlight. Summing up, he says he finds that the variation of capacity is dependent upon the absolute height above sea level, though in a small degree; on the relative height of the conducting surface or capacity with respect to the bodies surrounding it; on the relative change of the circuit with respect to the sun, caused by diurnal rotation of the earth; and—on the distance of the earth from the sun! He considers it probable that instruments will be perfected for indicating the altitude of a place by means of a circuit properly constructed and arranged, and adds that he has thought of a number of uses to which this principle may be put. It is also said that inductance is seemingly modified by the same causes that tend to modify the capacity. Although no reference is made to the Pupin theory concerning the use of inductance in overcoming the disturbing effects of the capacity of telephone lines, the "discovery" appears to invalidate that theory.